Semester	B.E. Semester VII – Computer Engineering
Subject	DevOps Lab
Subject In-charge	Dr. Snehal Andhare

Student Name	Vicky Pukale
Roll Number	23102A0033
Grade and Subject Teacher's Signature	

Task Number	13
Project Title:	Secure CI/CD Pipeline for a Loan Application Tracking Portal
Task Title	Configuration Management Script: Identify all server prerequisites for the Loan Application Tracking Portal: packages, users, folders, files, ports and services. Create either a Puppet manifest/modules or an Ansible inventory and YAML playbook to configure the target node.
Deliverable:	

1. How Ansible Works in This Project

Ansible operates **agentlessly** over standard SSH or local connection. It executes tasks only when changes are needed, guaranteeing **idempotency** (running the playbook once or a hundred times yields the exact same stable state without side effects).

2. Step-by-Step Commands to Run the Playbook

1: Verify Playbook Syntax

Navigate to the `ansible/` folder and run:

```
bash
cd ansible
ansible-playbook -i inventory.ini playbook.yml --syntax-check
```

```
D:\Vicky_Project\DevOps\DevOps_Project\ansible>docker run --rm -v "%cd%":/ansible my-ansible -i inventory.ini playbook.yml --syntax-check

playbook: playbook.yml
```

2: First Execution / Simulation Log (Dry-Run Mode)

```
D:\Vicky_Project\DevOps\DevOps_Project\ansible>docker run --rm -v "%cd%":/ansible my-ansible -i inventory.ini playbook.yml --check

PLAY [Provision Target Node for Loan Application Tracking Portal] **************

TASK [Gathering Facts] *********************************************************
ok: [localhost]

TASK [Update APT package cache] ************************************************
ok: [localhost]

TASK [Install core system packages and runtimes (Java 17, Docker, UFW)] ********
changed: [localhost]

TASK [Ensure Docker service is started and enabled on boot] ********************
skipping: [localhost]

TASK [Create dedicated application group: loanapp] *****************************
changed: [localhost]

TASK [Create unprivileged application user: loanapp] ***************************
changed: [localhost]

TASK [Create application directory hierarchy under /opt/loan-portal] ***********
[WARNING]: failed to look up user loanapp. Create user up to this point in real play
[WARNING]: failed to look up group loanapp. Create group up to this point in real play
changed: [localhost] => (item={'path': '/opt/loan-portal', 'mode': '0755'})
changed: [localhost] => (item={'path': '/opt/loan-portal/logs', 'mode': '0755'})
changed: [localhost] => (item={'path': '/opt/loan-portal/config', 'mode': '0750'})
changed: [localhost] => (item={'path': '/opt/loan-portal/data', 'mode': '0750'})

TASK [Deploy application environment configuration file] ***********************
changed: [localhost]

TASK [Allow SSH inbound traffic (Port 22)] *************************************
skipping: [localhost]

TASK [Allow Loan Application Tracking Portal inbound traffic (Port 8081)] ******
skipping: [localhost]

TASK [Verify Docker engine response] *******************************************
skipping: [localhost]

TASK [Verify Java 17 runtime availability] *************************************
skipping: [localhost]

TASK [Display provisioning summary] ********************************************
ok: [localhost] => {
    "msg": [
        "============================================================",
        "✅ NODE PROVISIONING COMPLETE & IDEMPOTENT!",
        "Target Host        : localhost",
        "Application User   : loanapp (UID: 1001)",
        "Target Directory   : /opt/loan-portal",
        "Configured Port    : 8081",
        "Docker Runtime     : ",
        "============================================================"
    ]
}

PLAY RECAP *********************************************************************
localhost                  : ok=8    changed=5    unreachable=0    failed=0    skipped=5    rescued=0    ignored=0
```

3. Commit Ansible Configuration to Git

In your Windows terminal:

```
cmd
git add ansible/ docs/SERVER_CONFIGURATION_SPECIFICATION.md
git commit -m "feat(ansible): add server configuration
specification, inventory, and automated provisioning playbook"
git push origin main
```

```
D:\Vicky_Project\DevOps\DevOps_Project>git add ansible/ docs/SERVER_CONFIGURATION_SPECIFICATION.md
warning: in the working copy of 'ansible/Dockerfile', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'ansible/ansible.cfg', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'ansible/inventory.ini', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'ansible/playbook.yml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'docs/SERVER_CONFIGURATION_SPECIFICATION.md', LF will be replaced by CRLF the next time Git touches it

D:\Vicky_Project\DevOps\DevOps_Project>git commit -m "feat(ansible): add server configuration specification, inventory, and automated provisioning playbook"

[main 7fd7657] feat(ansible): add server configuration specification, inventory, and automated provisioning playbook
 5 files changed, 238 insertions(+)
 create mode 100644 ansible/Dockerfile
 create mode 100644 ansible/ansible.cfg
 create mode 100644 ansible/inventory.ini
 create mode 100644 ansible/playbook.yml
 create mode 100644 docs/SERVER_CONFIGURATION_SPECIFICATION.md

D:\Vicky_Project\DevOps\DevOps_Project>git push origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 16 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (9/9), 3.79 KiB | 1.26 MiB/s, done.
Total 9 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/vickypukale/loan-application-tracking-portal.git
   bc27765..7fd7657  main -> main
```